

Supplementary Material II: Data Augmentation Ablation Studies

Overview

This supplementary material presents a comprehensive ablation study on the effect of synthetic data augmentation on dental caries segmentation. We evaluate seven encoder-decoder architectures—U-Net, U-Net++, FPN, LinkNet, MA-Net, PAN, and PSPNet—each trained under five augmentation configurations. The augmentation level is controlled by a multiplier: Aug1 corresponds to the original training set only ($1\times$), while Aug2 through Aug5 progressively increase the training set to $2\times$, $3\times$, $4\times$, and $5\times$ by sampling from a pool of synthetically generated image-mask pairs. All models use a ResNet-34 encoder initialised with ImageNet weights. Performance is evaluated on a held-out test set using per-class Dice and IoU scores at the best-performing epoch during training. The six evaluation classes correspond to single-caries categories—Simple Caries (SC), Medium Caries (MC), and Deep Caries (DC)—as well as their clinical combinations: MC+SC, MC+DC, and SC+MC+DC.

1 Effect of Augmentation Level on Overall Performance

Table 1 and Table 2 report the mean Dice and IoU scores, respectively, across all six classes for every model-augmentation combination. The optimal augmentation level varies across architectures, which is an important observation. Four of the seven models—U-Net++, FPN, LinkNet, and MA-Net—achieve their best average performance at Aug4 or Aug5, suggesting that these architectures benefit from substantial increases in training data volume. U-Net, interestingly, peaks at Aug3 (mean Dice 0.409, mean IoU 0.277), beyond which additional synthetic samples provide diminishing or even negative returns. PSPNet shows a similar pattern, with its best Dice at Aug5 (0.389) but its best IoU at Aug2 (0.263), indicating that the ranking can shift depending on the evaluation metric.

The most striking outlier is PAN, which achieves its best performance under Aug1—the baseline configuration with no synthetic augmentation (Dice 0.404, IoU 0.271). This stands in contrast to every other architecture, all of which benefit to some degree from augmented training data. One plausible explanation is that PAN’s pixel attention mechanism already provides strong implicit regularisation, and the addition of synthetic samples may introduce distributional shifts that interfere with its learned attention maps. This finding underscores the importance of architecture-specific augmentation tuning rather than applying a universal augmentation multiplier.

Across all configurations, U-Net++ with Aug4 achieves the highest overall performance on both metrics (Dice 0.437, IoU 0.297), followed by LinkNet with Aug5 (Dice 0.423, IoU 0.287). The gap between the best and worst model-augmentation pairings is substantial: U-Net++ at Aug4 outperforms FPN at Aug1 by 0.096 in Dice and 0.072 in IoU, highlighting that the combined choice of architecture and augmentation level can profoundly influence segmentation quality.

Table 1: Mean Dice score across all classes for each model–augmentation combination. **Bold** marks the best augmentation for each model.

Model	Aug1	Aug2	Aug3	Aug4	Aug5
U-Net	0.395	0.392	0.409	0.396	0.406
U-Net++	0.373	0.397	0.393	0.437	0.408
FPN	0.341	0.372	0.375	0.372	0.396
LinkNet	0.372	0.397	0.395	0.375	0.423
MA-Net	0.380	0.384	0.372	0.405	0.383
PAN	0.404	0.389	0.384	0.364	0.378
PSPNet	0.348	0.388	0.363	0.385	0.389

Table 2: Mean IoU score across all classes for each model–augmentation combination. **Bold** marks the best augmentation for each model.

Model	Aug1	Aug2	Aug3	Aug4	Aug5
U-Net	0.262	0.258	0.277	0.263	0.273
U-Net++	0.247	0.264	0.267	0.297	0.274
FPN	0.225	0.249	0.249	0.248	0.262
LinkNet	0.246	0.266	0.266	0.248	0.287
MA-Net	0.254	0.254	0.245	0.270	0.254
PAN	0.271	0.258	0.260	0.239	0.252
PSPNet	0.230	0.263	0.237	0.258	0.257

2 Detailed Per-Class Breakdown

We now provide the full per-class results at each augmentation level. Tables 3–4 report Dice scores and Tables 5–6 report IoU scores. These are split across two tables per metric for readability. Bold entries denote the best score for a given class across *all* 35 model–augmentation combinations.

A consistent pattern that emerges from the per-class data is the hierarchy of difficulty among the segmentation targets. Deep Caries (DC) is the most reliably segmented class across all experimental conditions, with Dice scores exceeding 0.44 in every configuration and reaching as high as 0.553 (PSPNet, Aug2). This is expected given that deep caries lesions tend to be larger and more visually distinct. At the other extreme, Simple Caries (SC) remains the most challenging category, with Dice scores rarely exceeding 0.23. The best SC segmentation is achieved by U-Net++ under Aug1 (Dice 0.245), though the margin over other competitive configurations is narrow. This difficulty is consistent with the clinical reality that early-stage caries are subtle, small, and frequently confused with healthy tissue even by human annotators.

Medium Caries (MC) occupies a middle ground, with scores that benefit substantially from augmentation. U-Net++ at Aug4 achieves the highest MC Dice of 0.345—a 51% relative improvement over its Aug1 score of 0.228—demonstrating that the augmented data is particularly beneficial for this intermediate class. A similar trend holds for the combined MC+SC class, where U-Net++ at Aug4 again leads with a Dice of 0.419 compared to its baseline of 0.310. These improvements suggest that the synthetic data successfully introduces additional positional and morphological variability for medium-severity lesions, which helps the model generalise beyond the limited texture patterns present in the original training set.

The multi-class combinations (MC+DC, SC+MC+DC) tend to show more stable improvements with augmentation compared to the single-class metrics. This is because the multi-class evaluation aggregates over larger foreground regions, smoothing out the noise inherent in evalu-

Table 3: Per-class Dice scores for U-Net, U-Net++, FPN, and LinkNet across all augmentation levels. **Bold** indicates the best score for each class across all 35 model–augmentation configurations.

Class	U-Net					U-Net++					FPN					LinkNet				
	Aug1	Aug2	Aug3	Aug4	Aug5	Aug1	Aug2	Aug3	Aug4	Aug5	Aug1	Aug2	Aug3	Aug4	Aug5	Aug1	Aug2	Aug3	Aug4	Aug5
SC	0.234	0.232	0.213	0.230	0.222	0.245	0.227	0.205	0.217	0.227	0.182	0.210	0.168	0.145	0.199	0.202	0.225	0.179	0.216	0.203
MC	0.279	0.296	0.283	0.295	0.312	0.228	0.311	0.263	0.345	0.320	0.226	0.191	0.248	0.273	0.322	0.249	0.284	0.284	0.246	0.330
DC	0.505	0.457	0.524	0.460	0.487	0.494	0.488	0.511	0.530	0.515	0.442	0.508	0.525	0.482	0.495	0.483	0.495	0.533	0.473	0.519
MC+SC	0.319	0.339	0.354	0.369	0.376	0.310	0.340	0.303	0.419	0.363	0.279	0.278	0.282	0.285	0.337	0.312	0.329	0.325	0.324	0.375
MC+DC	0.522	0.521	0.542	0.509	0.519	0.481	0.510	0.545	0.552	0.512	0.460	0.524	0.519	0.532	0.516	0.494	0.528	0.529	0.498	0.560
SC+MC+DC	0.512	0.509	0.541	0.513	0.521	0.480	0.502	0.530	0.561	0.511	0.460	0.520	0.507	0.514	0.506	0.489	0.519	0.518	0.497	0.552
Average	0.395	0.392	0.409	0.396	0.406	0.373	0.397	0.393	0.437	0.408	0.341	0.372	0.375	0.372	0.396	0.372	0.397	0.395	0.375	0.423

Table 4: Per-class Dice scores for MA-Net, PAN, and PSPNet across all augmentation levels. **Bold** indicates the best score for each class across all 35 model–augmentation configurations.

Class	MA-Net					PAN					PSPNet				
	Aug1	Aug2	Aug3	Aug4	Aug5	Aug1	Aug2	Aug3	Aug4	Aug5	Aug1	Aug2	Aug3	Aug4	Aug5
SC	0.184	0.205	0.189	0.230	0.222	0.225	0.178	0.208	0.187	0.159	0.118	0.181	0.168	0.169	0.180
MC	0.282	0.282	0.266	0.334	0.287	0.258	0.286	0.258	0.231	0.278	0.189	0.253	0.263	0.270	0.302
DC	0.492	0.488	0.471	0.470	0.445	0.538	0.499	0.486	0.443	0.484	0.536	0.553	0.479	0.502	0.487
MC+SC	0.307	0.316	0.318	0.380	0.339	0.335	0.324	0.327	0.349	0.321	0.226	0.268	0.277	0.304	0.324
MC+DC	0.514	0.509	0.494	0.511	0.504	0.537	0.527	0.512	0.477	0.521	0.517	0.552	0.503	0.542	0.525
SC+MC+DC	0.502	0.502	0.492	0.508	0.503	0.532	0.523	0.512	0.496	0.508	0.500	0.525	0.489	0.526	0.513
Average	0.380	0.384	0.372	0.405	0.383	0.404	0.389	0.384	0.364	0.378	0.348	0.388	0.363	0.385	0.389

ating small, rare structures. LinkNet at Aug5 achieves the highest MC+DC Dice of 0.560, while U-Net++ at Aug4 leads on SC+MC+DC with 0.561. These two architectures, both of which incorporate skip connections or dense pathways, appear to benefit the most from augmentation at the multi-class level.

3 Best-Configuration Comparison Across Architectures

To distil the ablation into a single cross-architecture comparison, Table 7 and Table 8 present the per-class scores obtained by each model under its individually optimal augmentation level.

U-Net++ with $4\times$ augmentation emerges as the strongest overall model, achieving the best scores on four of the six class-level metrics in Dice (MC, MC+SC, SC+MC+DC, and the overall average) and three in IoU (MC, MC+SC, SC+MC+DC). Its average Dice of 0.437 and IoU of 0.297 represent clear margins over the second-best configuration (LinkNet with Aug5, at 0.423 Dice and 0.287 IoU). The dense nested skip connections in U-Net++ appear to synergise well with the additional training diversity provided by synthetic augmentation, enabling it to capture both fine-grained and coarse-grained caries morphologies.

LinkNet, despite its comparatively lightweight decoder, achieves the best MC+DC Dice (0.560) and IoU (0.401) under $5\times$ augmentation. This suggests that the efficient residual-based decoder can still achieve competitive performance when given sufficient training diversity, particularly for the more prevalent caries classes. FPN, which shares a similar philosophy of multi-scale feature aggregation, benefits most at $5\times$ augmentation as well, though its absolute scores remain lower than LinkNet’s across all classes.

For Deep Caries specifically, PAN under Aug1 (Dice 0.538) and PSPNet under Aug2 (Dice 0.553) outperform all other configurations. This class, being the largest and most visually distinct, does not require augmentation to achieve reasonable segmentation. In fact, several models exhibit a decline in DC performance as augmentation increases (e.g., MA-Net drops from 0.492 at Aug1 to 0.445 at Aug5), suggesting that the synthetic data may dilute the already-strong signal for this class. The pattern is reversed for harder classes such as MC and MC+SC, where nearly all models show monotonic improvement with augmentation.

Table 5: Per-class IoU scores for U-Net, U-Net++, FPN, and LinkNet across all augmentation levels. **Bold** indicates the best score for each class across all 35 model–augmentation configurations.

Class	U-Net					U-Net++					FPN					LinkNet				
	Aug1	Aug2	Aug3	Aug4	Aug5	Aug1	Aug2	Aug3	Aug4	Aug5	Aug1	Aug2	Aug3	Aug4	Aug5	Aug1	Aug2	Aug3	Aug4	Aug5
SC	0.137	0.136	0.123	0.138	0.128	0.143	0.133	0.117	0.125	0.153	0.103	0.121	0.098	0.082	0.112	0.116	0.131	0.104	0.128	0.118
MC	0.177	0.182	0.173	0.180	0.199	0.134	0.198	0.167	0.218	0.178	0.135	0.114	0.148	0.166	0.201	0.150	0.176	0.184	0.148	0.206
DC	0.350	0.313	0.378	0.318	0.341	0.357	0.342	0.372	0.376	0.352	0.315	0.365	0.368	0.335	0.344	0.343	0.355	0.378	0.327	0.372
MC+SC	0.195	0.207	0.221	0.231	0.236	0.186	0.213	0.184	0.270	0.233	0.166	0.167	0.168	0.171	0.207	0.192	0.201	0.201	0.198	0.235
MC+DC	0.362	0.360	0.385	0.354	0.367	0.332	0.353	0.389	0.393	0.363	0.318	0.368	0.361	0.375	0.358	0.343	0.371	0.370	0.346	0.401
SC+MC+DC	0.350	0.348	0.381	0.356	0.365	0.328	0.343	0.372	0.398	0.365	0.313	0.360	0.349	0.358	0.347	0.335	0.360	0.358	0.342	0.391
Average	0.262	0.258	0.277	0.263	0.273	0.247	0.264	0.267	0.297	0.274	0.225	0.249	0.249	0.248	0.262	0.246	0.266	0.266	0.248	0.287

Table 6: Per-class IoU scores for MA-Net, PAN, and PSPNet across all augmentation levels. **Bold** indicates the best score for each class across all 35 model–augmentation configurations.

Class	MA-Net					PAN					PSPNet				
	Aug1	Aug2	Aug3	Aug4	Aug5	Aug1	Aug2	Aug3	Aug4	Aug5	Aug1	Aug2	Aug3	Aug4	Aug5
SC	0.106	0.118	0.108	0.132	0.131	0.132	0.101	0.079	0.106	0.083	0.066	0.105	0.094	0.096	0.103
MC	0.174	0.173	0.160	0.208	0.162	0.162	0.172	0.170	0.134	0.113	0.110	0.158	0.161	0.165	0.187
DC	0.349	0.339	0.330	0.331	0.317	0.380	0.346	0.375	0.308	0.384	0.377	0.399	0.325	0.353	0.336
MC+SC	0.187	0.193	0.193	0.239	0.194	0.208	0.198	0.179	0.214	0.157	0.132	0.159	0.167	0.185	0.197
MC+DC	0.359	0.354	0.341	0.358	0.366	0.377	0.367	0.387	0.329	0.395	0.356	0.394	0.344	0.382	0.367
SC+MC+DC	0.346	0.346	0.336	0.353	0.355	0.369	0.362	0.370	0.341	0.381	0.339	0.365	0.330	0.366	0.354
Average	0.254	0.254	0.245	0.270	0.254	0.271	0.258	0.260	0.239	0.252	0.230	0.263	0.237	0.258	0.257

4 Cross-Cutting Analysis

4.1 Augmentation Sensitivity Across Architectures

The variance in augmentation response across architectures reveals a fundamental difference in how each model leverages additional training data. We define the *augmentation sensitivity* of a model as the range (maximum minus minimum) of its mean Dice scores across the five augmentation levels. By this measure, U-Net++ exhibits the highest sensitivity (range: $0.437 - 0.373 = 0.064$), followed by FPN (0.055) and PSPNet (0.041). U-Net shows a comparatively modest range of 0.017, indicating that its performance is relatively stable regardless of how much synthetic data is added. PAN shows a range of 0.040, but uniquely in the negative direction—its performance degrades monotonically from Aug1 to Aug4 before recovering slightly at Aug5. This reversed sensitivity makes PAN particularly unsuitable for naïve augmentation strategies.

The practical implication is that architecture selection and augmentation tuning should not be treated as independent decisions. A researcher choosing FPN for its computational efficiency, for example, would need to use $5\times$ augmentation to approach the performance of U-Net or PAN at baseline. Conversely, U-Net++ at $4\times$ augmentation would be the recommended configuration when the goal is to maximise absolute performance, at the cost of a more complex architecture and larger training set.

4.2 Class-Specific Augmentation Interactions

The interaction between augmentation level and class difficulty reveals a nuanced picture. For the hardest class (SC), most models show either flat or declining performance beyond Aug2—U-Net drops from 0.234 (Aug1) to 0.213 (Aug3), and PAN drops from 0.225 (Aug1) to 0.159 (Aug5). The synthetic data, while expanding the overall training distribution, may not adequately represent the subtle visual characteristics of early-stage caries, leading to a form of negative transfer for this class. MA-Net is a notable exception, improving its SC Dice from 0.184 (Aug1) to 0.230 (Aug4), possibly reflecting the advantage of its multi-scale attention mechanism in leveraging diverse training samples for fine-grained structures.

For Medium Caries, the augmentation benefit is far more consistent. Five of the seven

Table 7: Per-class Dice scores at each model’s best augmentation level. **Bold** marks the best across all models for each class.

Class	U-Net (Aug3)	U-Net++ (Aug4)	FPN (Aug5)	LinkNet (Aug5)	MA-Net (Aug4)	PAN (Aug1)	PSPNet (Aug5)
SC	0.213	0.217	0.199	0.203	0.230	0.225	0.180
MC	0.283	0.345	0.322	0.330	0.334	0.258	0.302
DC	0.524	0.530	0.495	0.519	0.470	0.538	0.487
MC+SC	0.354	0.419	0.337	0.375	0.380	0.335	0.324
MC+DC	0.542	0.552	0.516	0.560	0.511	0.537	0.525
SC+MC+DC	0.541	0.561	0.506	0.552	0.508	0.532	0.513
Average	0.409	0.437	0.396	0.423	0.405	0.404	0.389

Table 8: Per-class IoU scores at each model’s best augmentation level. **Bold** marks the best across all models for each class.

Class	U-Net (Aug3)	U-Net++ (Aug4)	FPN (Aug5)	LinkNet (Aug5)	MA-Net (Aug4)	PAN (Aug1)	PSPNet (Aug2)
SC	0.123	0.125	0.112	0.118	0.132	0.132	0.105
MC	0.173	0.218	0.201	0.206	0.208	0.162	0.158
DC	0.378	0.376	0.344	0.372	0.331	0.380	0.399
MC+SC	0.221	0.270	0.207	0.235	0.239	0.208	0.159
MC+DC	0.385	0.393	0.358	0.401	0.358	0.377	0.394
SC+MC+DC	0.381	0.398	0.347	0.391	0.353	0.369	0.365
Average	0.277	0.297	0.262	0.287	0.270	0.271	0.263

models achieve their best MC score at Aug4 or Aug5: U-Net++ (0.345 at Aug4), MA-Net (0.334 at Aug4), LinkNet (0.330 at Aug5), FPN (0.322 at Aug5), and U-Net (0.312 at Aug5). The relative improvements over the baseline range from 10% (U-Net) to 51% (U-Net++). This suggests that medium-severity caries represent a class where the model has enough signal to learn from but insufficient training examples to generalise properly—exactly the regime where data augmentation is expected to yield the greatest benefit.

Deep Caries, as the easiest class, shows the most mixed augmentation response. Several architectures (MA-Net, PAN, U-Net) achieve their best DC score at Aug1, while others (FPN, LinkNet, U-Net++) perform better with moderate augmentation. The overall spread in DC Dice is narrow (0.443 to 0.553), confirming that this class can be reliably segmented even with limited data. The marginal value of augmentation for DC is therefore low, and the primary utility of synthetic data lies in improving performance on the lower-prevalence caries classes.

4.3 Architecture–Augmentation Trade-offs

When considering models under their respective best augmentation configurations, the ranking by average Dice is: U-Net++ (0.437) > LinkNet (0.423) > U-Net (0.409) > MA-Net (0.405) > PAN (0.404) > FPN (0.396) > PSPNet (0.389). The same ordering is largely preserved for IoU, with the exception that PAN and MA-Net swap positions. This stability across metrics supports the conclusion that U-Net++ and LinkNet genuinely capture the data distribution better when paired with the right augmentation, rather than benefiting from metric-specific artefacts.

It is worth noting that the performance gap between the top and bottom architectures (0.048

Dice, or about 12% relative) is of a comparable magnitude to the gap between the worst and best augmentation levels within a single architecture. For instance, FPN’s range across augmentation levels spans 0.055 Dice, nearly matching the inter-model spread. This highlights that the choice of augmentation configuration can be as consequential as the choice of architecture itself—a finding with practical implications for medical image analysis, where training data is scarce and augmentation is one of the few levers available to improve performance.